

SKYGUARD

Multimodal Failure Prognosis for UAV Delivery Fleet

Team Suui

Neil Mittal

Mahin Mandava

Roumya Lenka

Tanmay Khandelwal

Korla Bharat

Aadya Sisodya



Tabular Modality

STARTING WITH TABULAR MODALITY BY ANALYSING PATTERNS AND NEW FEATURES

Feature Engineering Strategy

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graph TD; A[Feature Engineering Strategy] --> B[Target Encoding]; A --> C[Ratio Normalization]; A --> D[Model Generalization];
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Target Encoding

- Applied localized target encoding (fuse technique) to categorical variables
- Effectively capturing label correlated variance without expanding the feature space.

Ratio Normalization

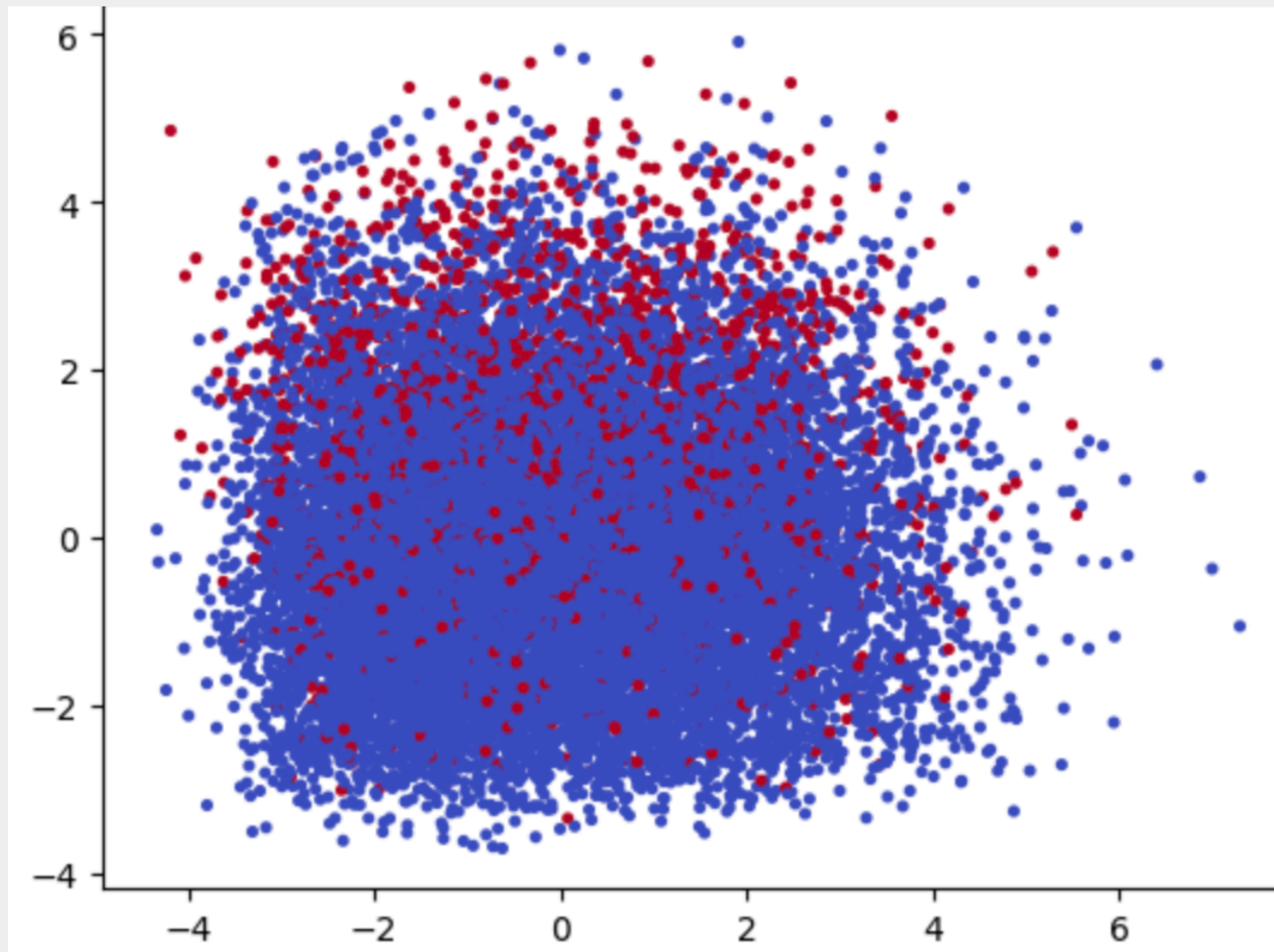
- Derived generalized features including Payload Ratio, Maneuvers per Minute
- Provides a general comparison rather than absolute terms

Model Generalization

- By comparing generalized relative readings rather than absolute unscaled values.
- We actively suppress overfitting and ensure robust performance on unseen datasets

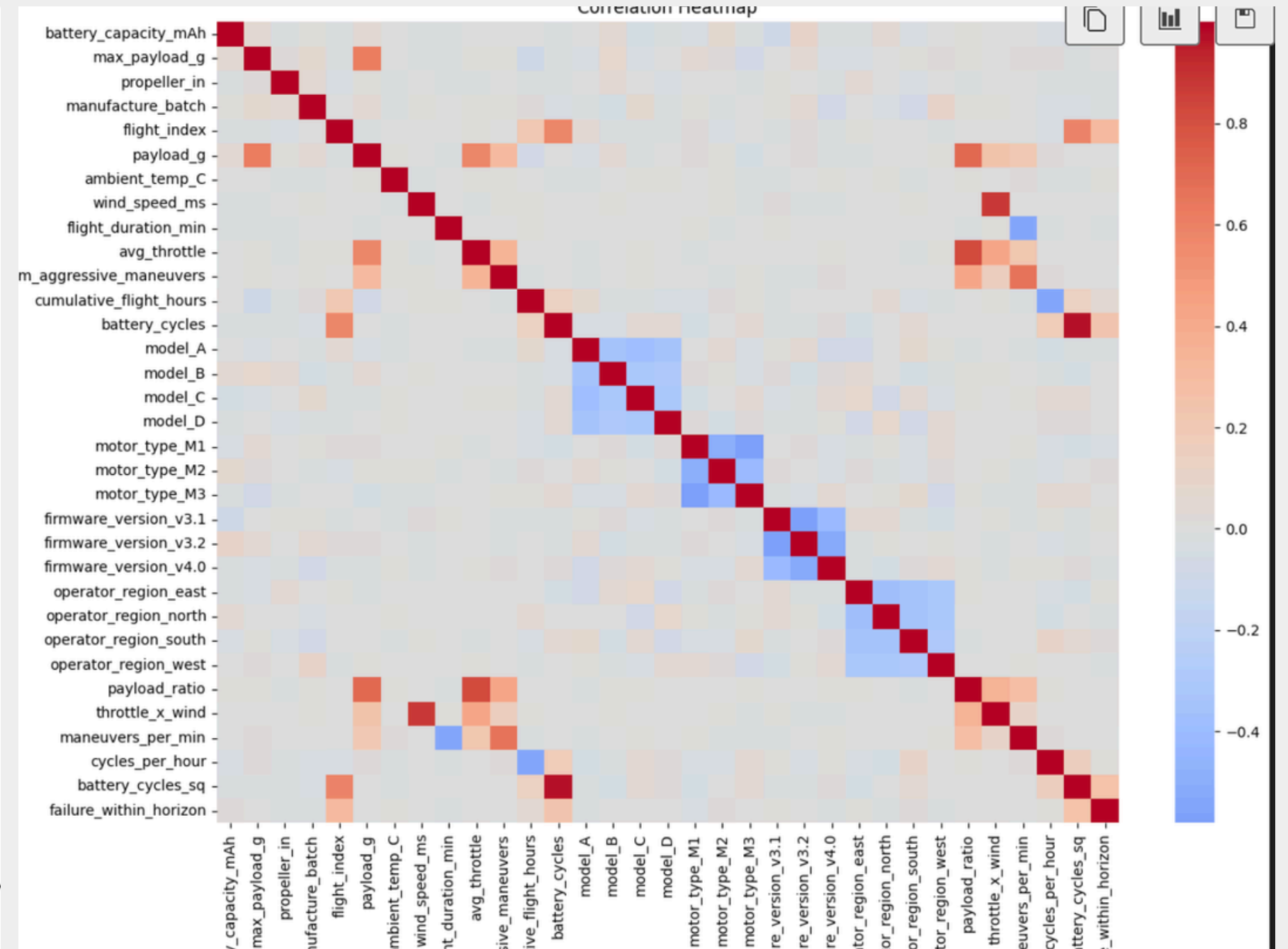
Tabular Modality

STARTING WITH TABULAR MODALITY BY ANALYSING PATTERNS AND NEW FEATURES



Principal Component Analysis (PCA)

2D projection reveals significant class overlap, indicating nonlinear decision boundaries will likely be required



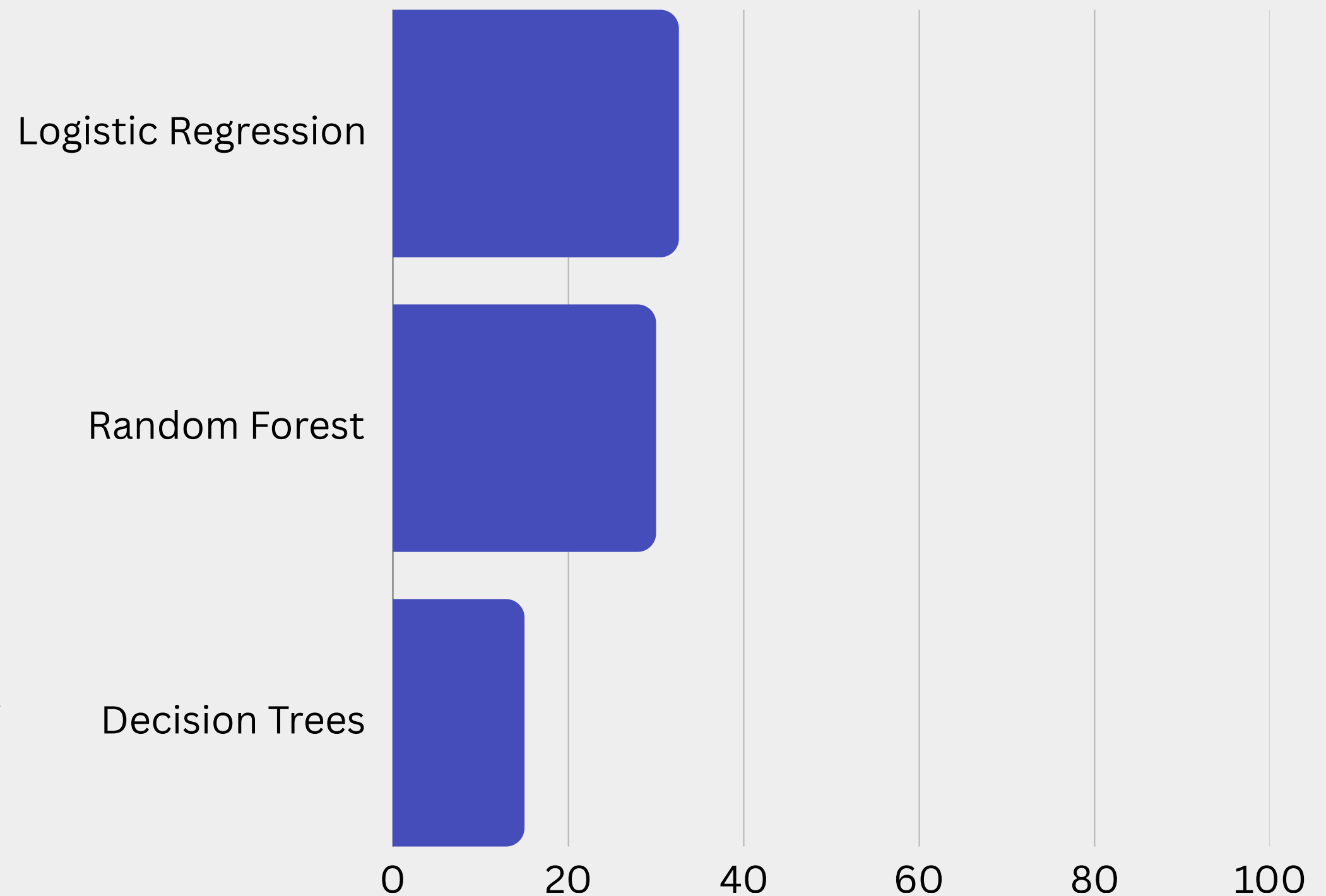
Feature Correlation Matrix

Heatmap identifies key positive and negative correlations, confirming the independence of our newly engineered ratio features

Baseline Model Evaluation

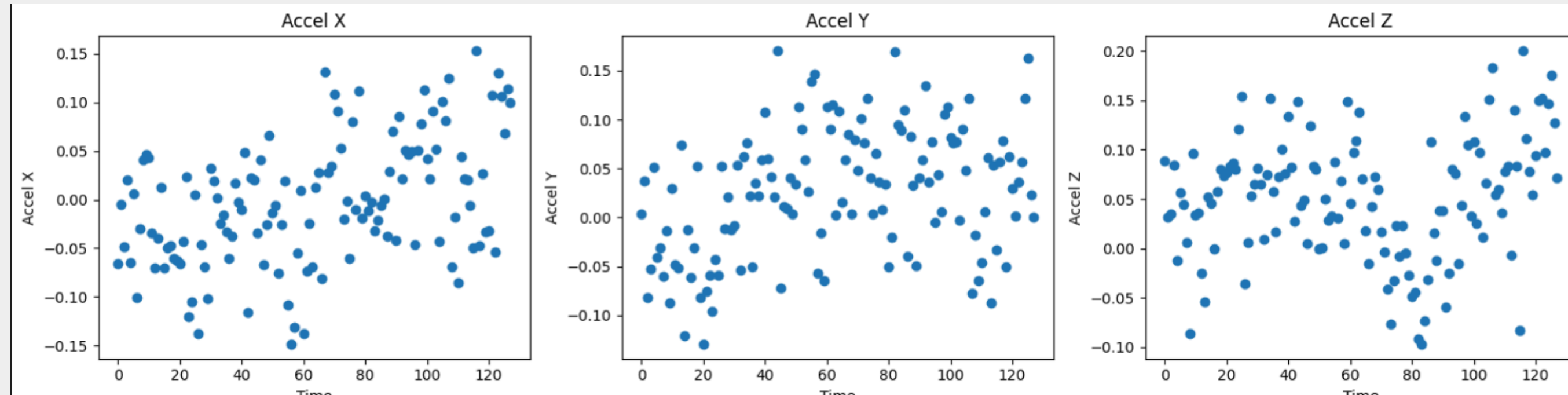
PERFORMANCE OF SIMPLER LINEAR MODELS OVER COMPLEX ENSEMBLES

- Logistic Regression sets current baseline ceiling at 32.6%. Very higher from low performing decision trees
- Interestingly, this simpler linear model slightly outperforms complex ensembles (Random Forest)
- Suggesting our feature engineering successfully linearized key relationships when combined together



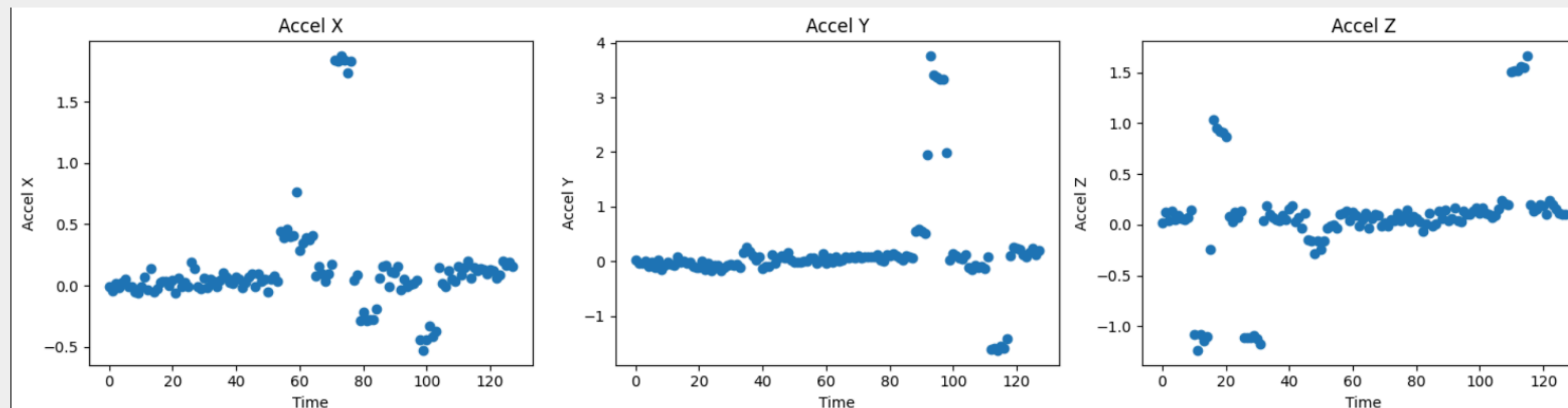
Signal Modality

SHIFTING TO SIGNAL MODALITY BY EVALUATING SHAPES AND FREQUENCY DOMAINS



Unhealthy Flight

- Raw accelerometer data exhibits high variance, chaotic dispersion
- Lacks a stable baseline across timeframes. The noise to signal ratio is visually prominent

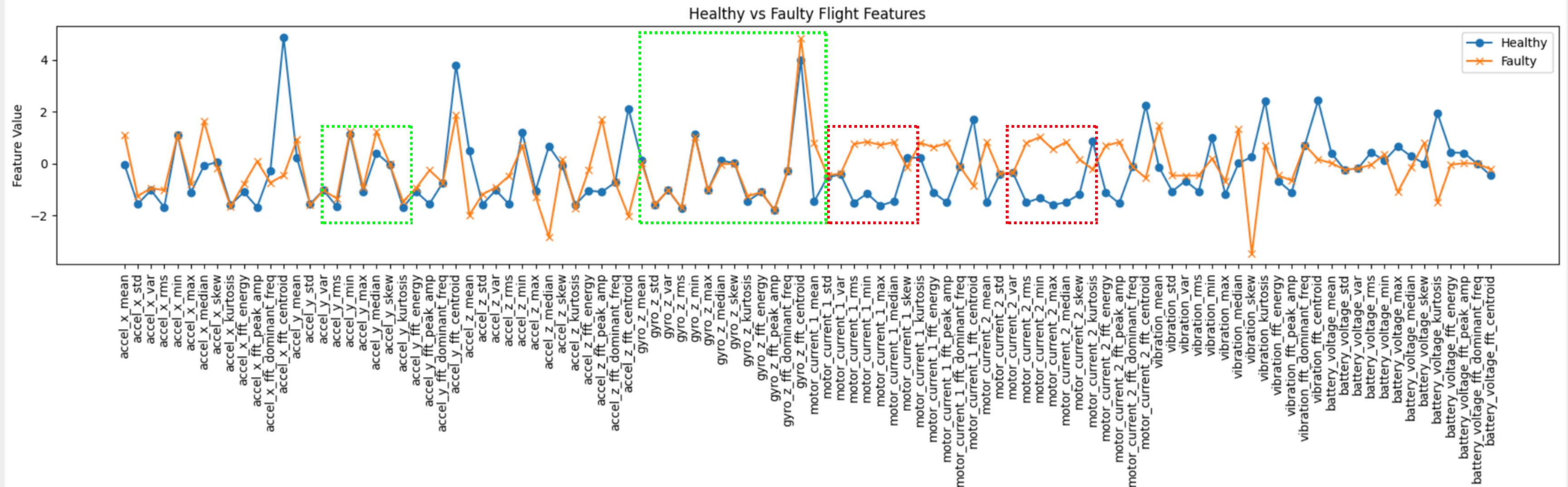


Healthy Flight

- Data points form tight, predictable clusters around a stable baseline (near zero)
- Expected maneuvers manifest as distinct, isolated spikes rather than continuous noise

Signal Modality

TRANSFORMING THE RAW TIMESERIES DATA INTO USEFUL FEATURES



Feature Engineering - Statistics

- Extracted core shape and volatility metrics including mean, median, var, rms, max, min, skew, kurtosis which provide much more information

Frequency Domain (FFT)

- Applied Fast Fourier Transforms to identify hidden oscillatory anomalies, extracting key features like Energy, Dominant Frequency, Centroid of Frequency

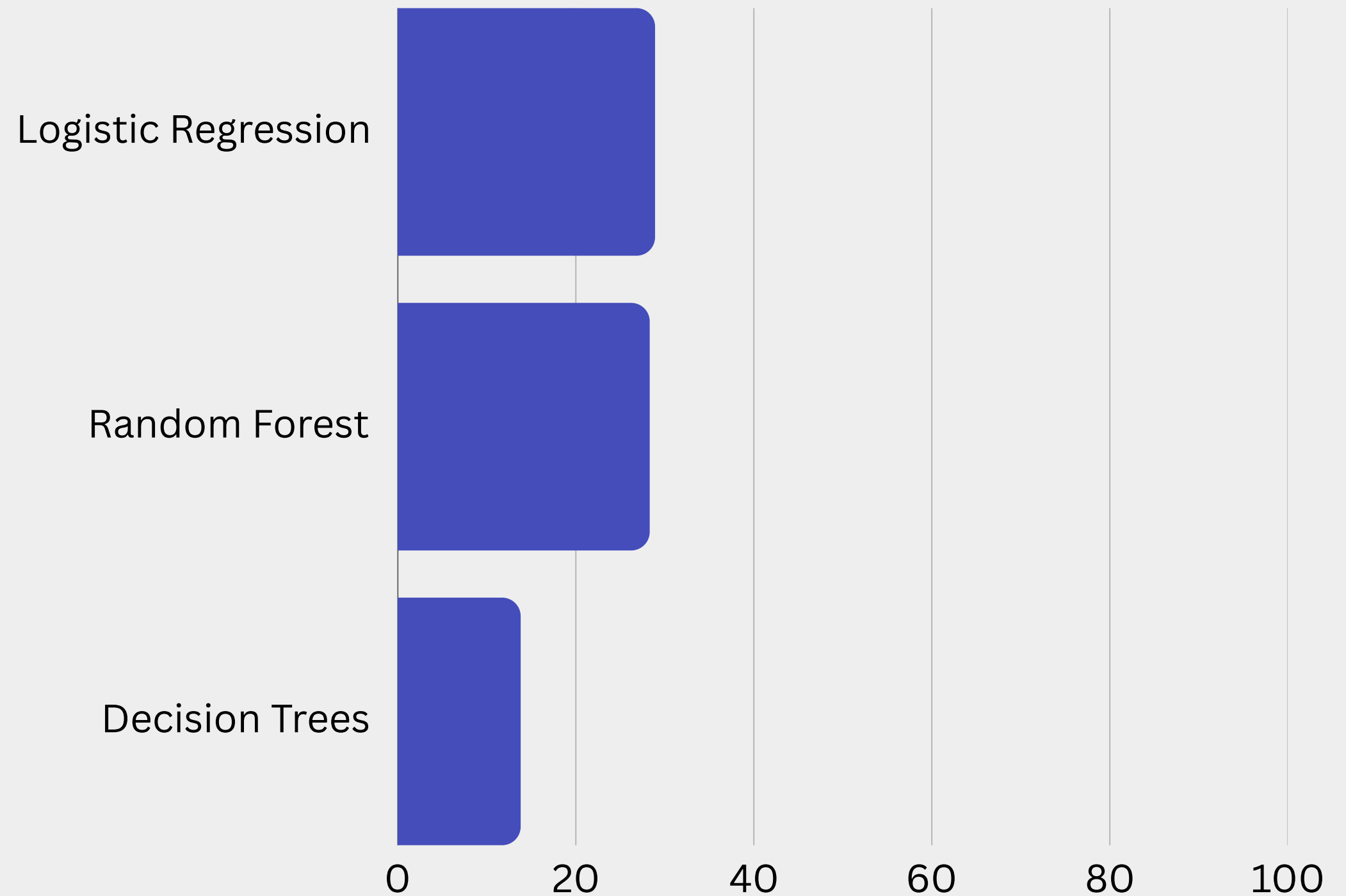
Feature Separability

- Visual analysis across these engineered dimensions confirms varying degrees of variance between classes, validating the need for multi-domain extraction

Baseline Model Evaluation

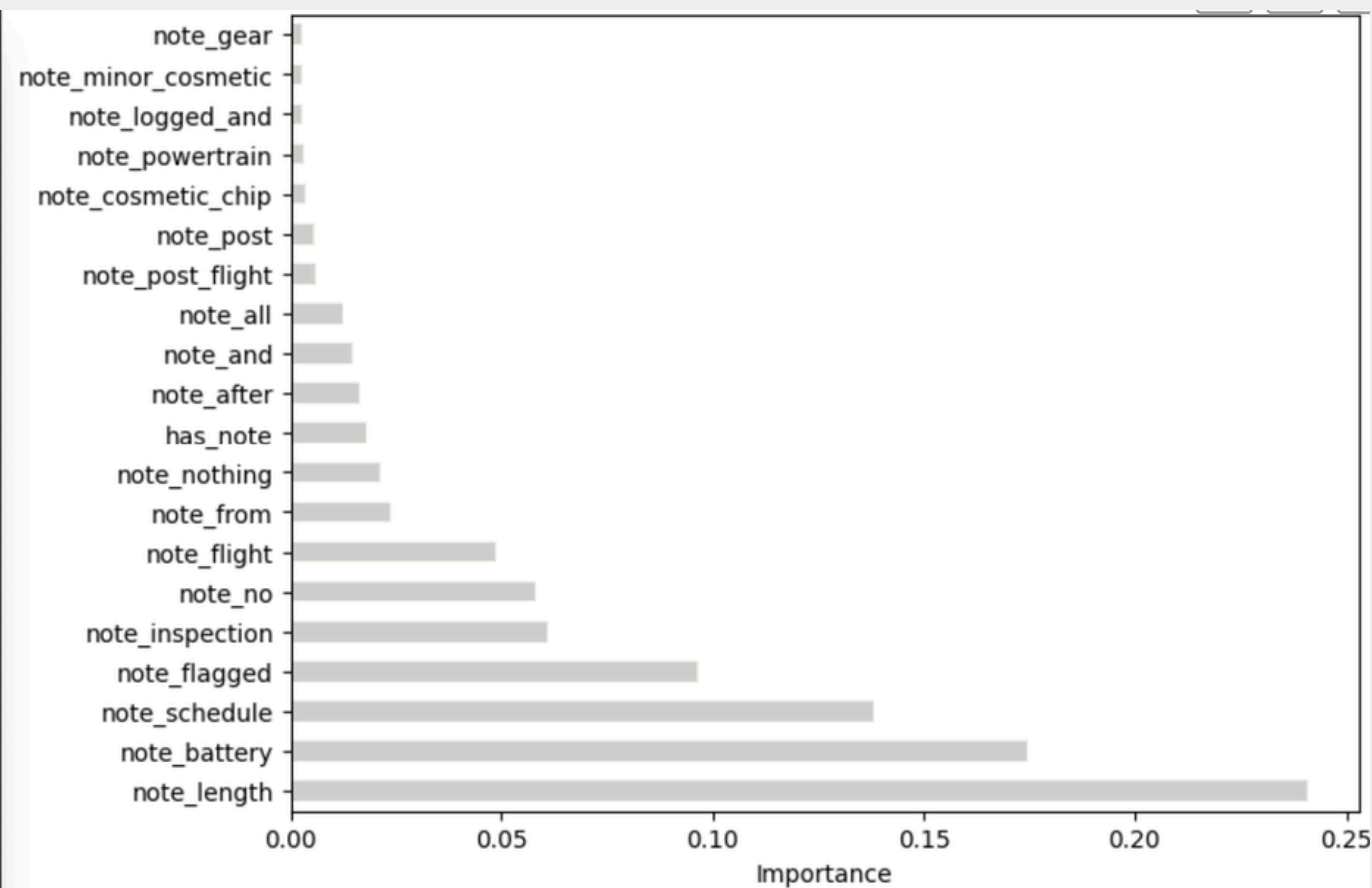
PERFORMANCE OF SIMPLER LINEAR MODELS OVER COMPLEX ENSEMBLES

- Logistic Regression and Random Forest perform almost identically on this feature set (~28%).
- Notably, this signal-based baseline underperforms the Tabular baseline (32.6%).
- Hypothesis: Statistical aggregation may be destroying important sequential context. Advanced time-series models.

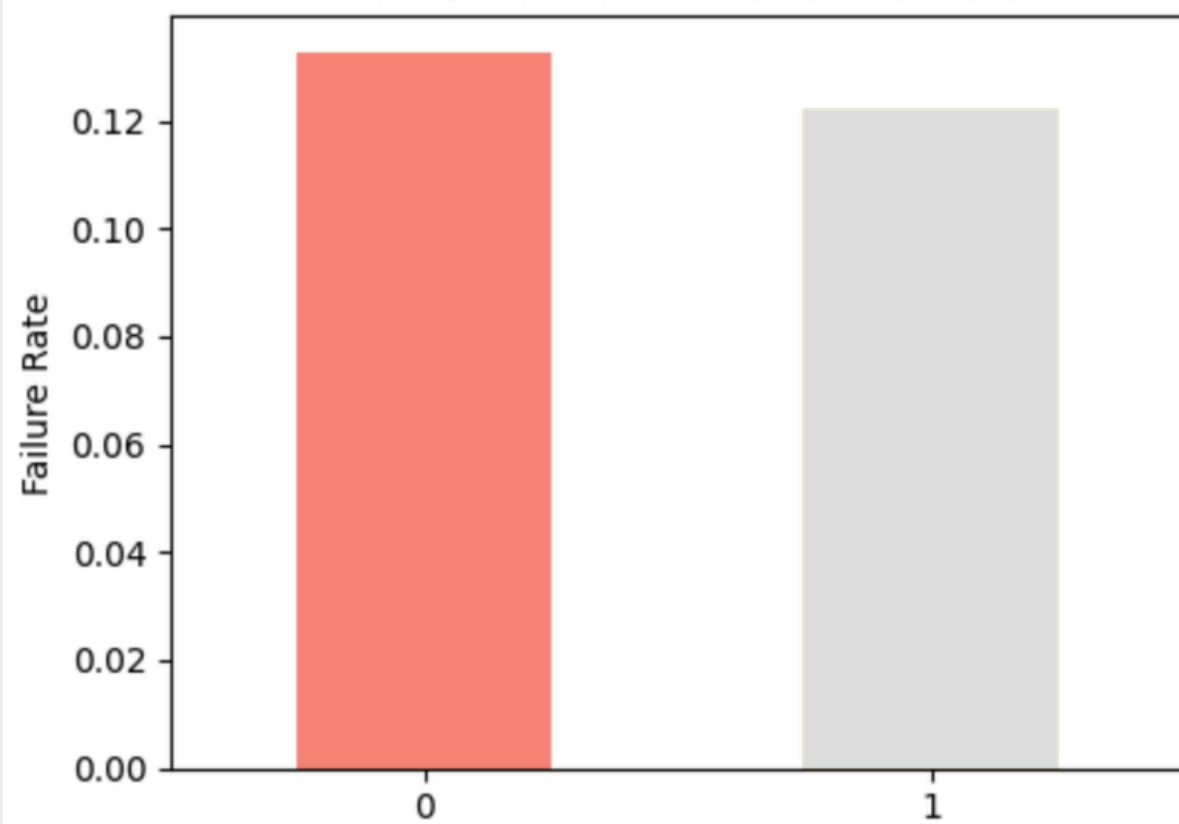


TF-IDF VECTORIZATION

VECTORIZING NOTES AND SEMANTIC FEATURES USING TF-IDF ALGORITHM



Failure Rate: No Note vs Has Note



Feature Importance

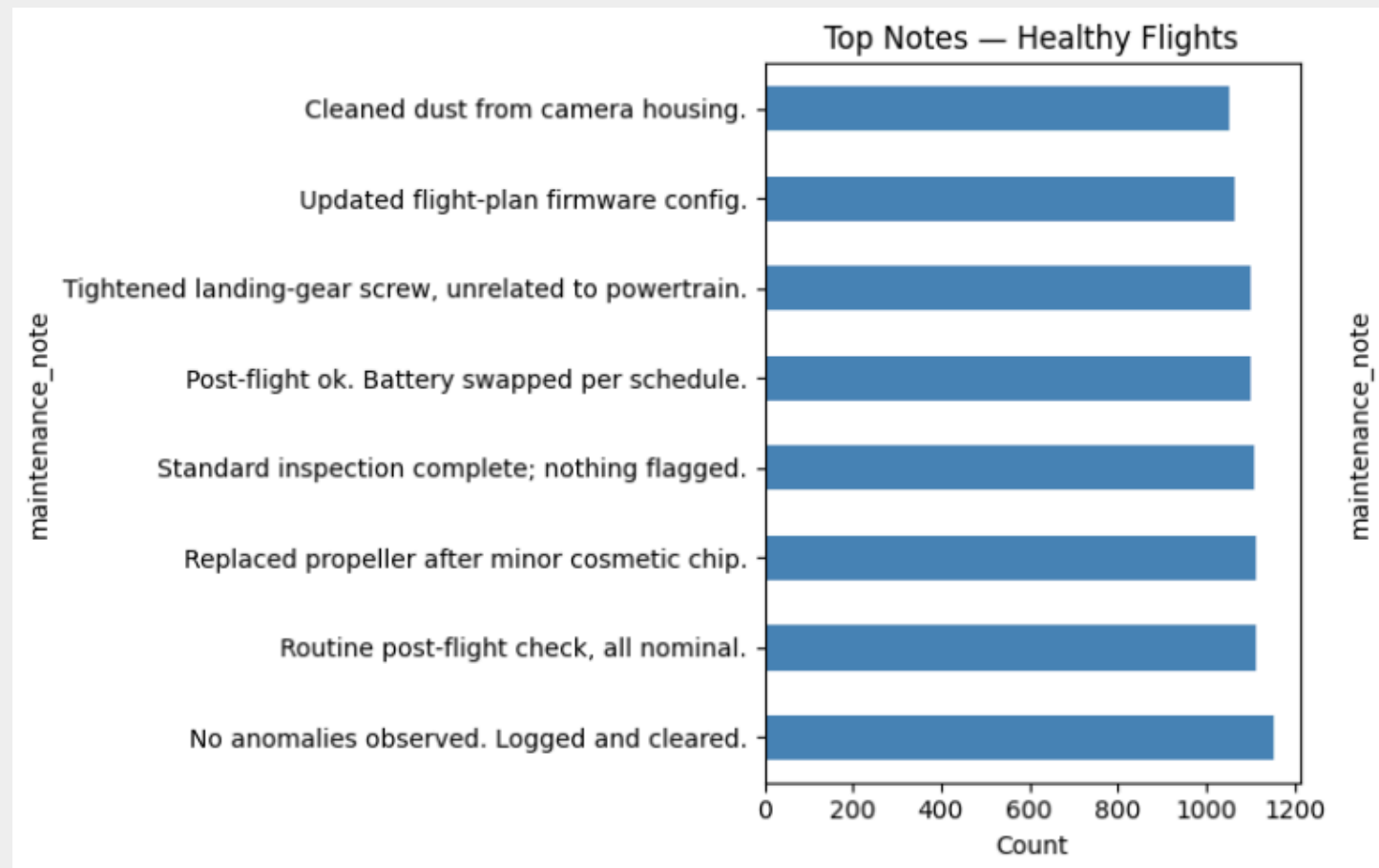
- Post TF-IDF vectorization, the length of the note emerges as the most dominant predictive feature
- Closely followed by specific operational keywords like battery and schedule and others.

The Has Note Insight

- We engineered a binary `has_note` feature. Surprisingly, flights without any recorded maintenance notes exhibit a slightly higher failure rate (~13%)
- Those with notes (~12%), indicating that routine, meticulous logging correlates with healthier overall airframes

TF-IDF VECTORIZATION

ANALYZING FURTHER PATTERNS IN THE NOTES DATASET TO IDENTIFY KEY ERRORS

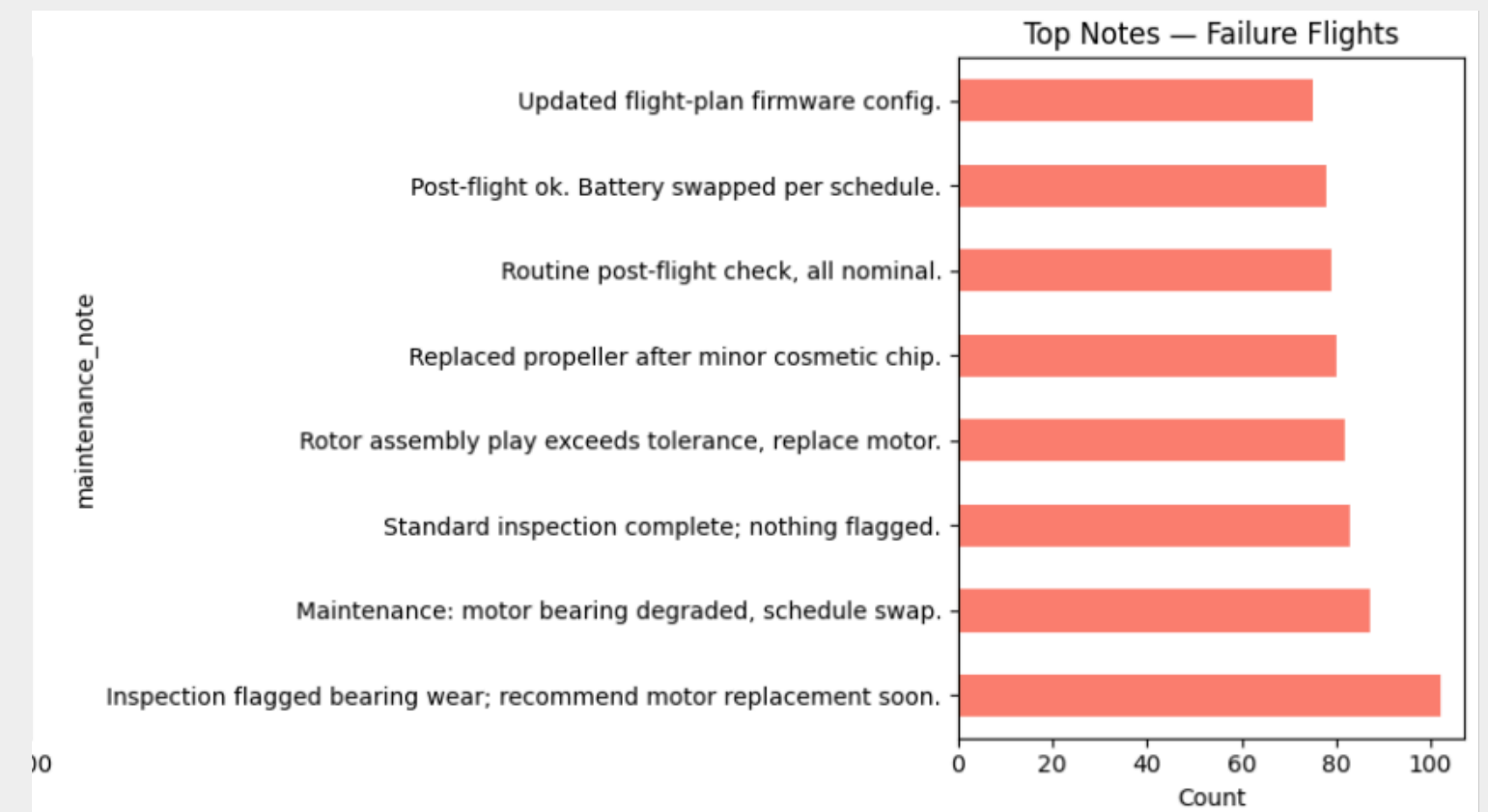


Failure Flights

- Reveals critical mechanical degradation terminology completely absent in healthy logs
- Signals proper examination with cautious notes of engineer may point to failure

Healthy Flights

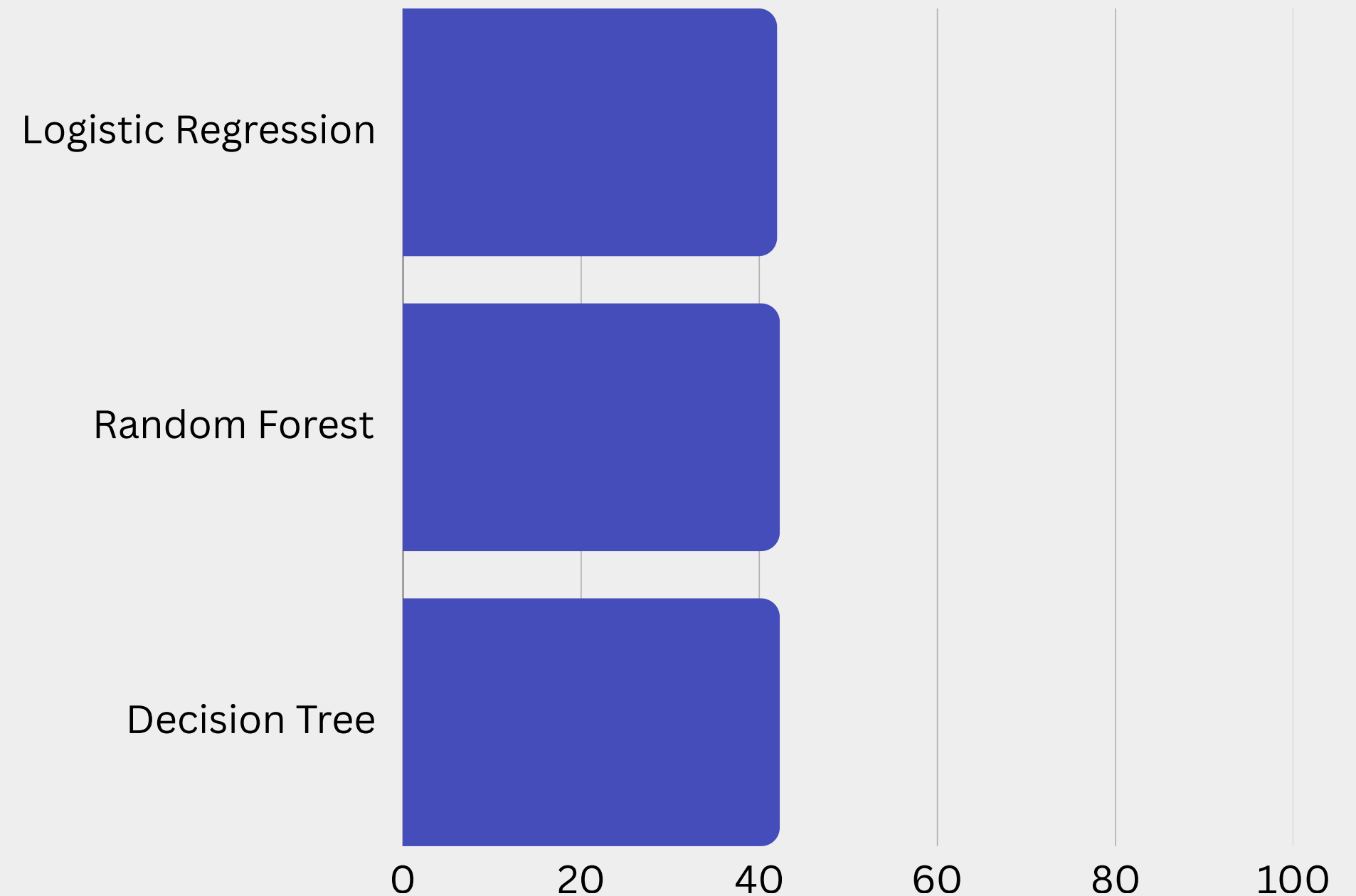
- Dominated by high-frequency, low severity routine confirmations
- Signals proper examination with healthy conditions results in healthy flights



TF-IDF VECTORIZATION

PERFORMANCE OF SIMPLER LINEAR MODELS OVER COMPLEX ENSEMBLES

- All perform almost identically on this feature set, with all three achieving an accuracy of approximately 42%.
- The simpler linear model (Logistic Regression) matches the performance of the complex ensemble methods.
- Core Hypothesis: Consequently, simpler linear models are entirely sufficient once the text data has been vectorized.



FUSING THE FINAL MODEL

MAKING THE FINAL PREDICTION BY FUSING ALL THREE MODALITIES

- **Data Fusion**

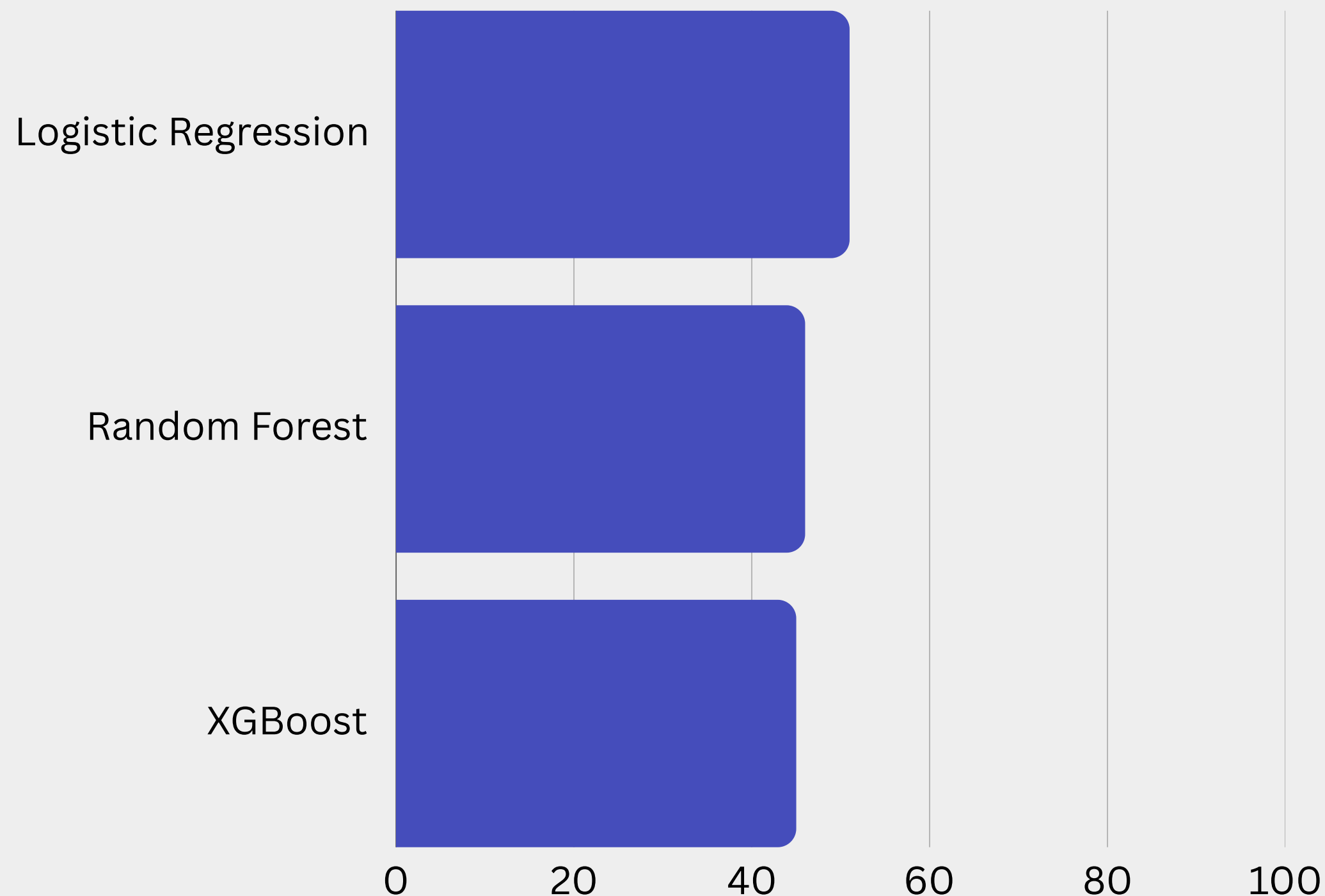
Concatenated all three engineered dataframes into a single feature space.

- **Target Encoding**

Applied to model_name instead of One-Hot to robustly handle the occurrence of unseen drone models during testing.

- **Robust Validation**

Implemented Group K-Fold split purely on drone_id to eliminate unit-specific data leakage.



Fly Longer and Stay in
Control Always



A drone built for stable signal
and extended daily flight.



THANK YOU